Symmetry-reduced variational quantum eigensolver captures the CO_2 dissociation curve with five qubits

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Abstract

Accurate potential-energy surfaces (PES) underpin catalyst design and carbon-capture research, yet the strongly correlated bond-breaking regime remains costly for conventional electronic-structure methods, e.g DFT. Quantum computing algorithms offer an alternative route, but their practical resource requirements for polyatomic molecules are still poorly characterised. Here we employ the variational quantum eigensolver (VQE) with a unitary coupled-cluster singles-and-doubles (UCCSD) ansatz to compute ground-state PES for H_2 and linear CO_2. For CO_2, a complete-active-space (CAS) reduction combined with symmetry-informed qubit tapering compresses the 30-spin-orbital problem to five qubits and 734 controlled-NOT gates, recovering 32% of the electron-correlation energy with less than 0.1% deviation from the full-configuration-interaction (FCI) limit. A systematic comparison of the Jordan–Wigner, Bravyi–Kitaev, and parity fermion-to-qubit mappings reveals that, after tapering, Jordan–Wigner compiles most efficiently for these small active spaces. Noise-model emulations using IBM device profiles quantify the energy penalties imposed by current

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hardware, establishing a concrete baseline for future error-mitigation studies on near-term quantum processors.

Keywords: quantum computing; variational quantum eigensolver; carbon dioxide; potential-energy surface; fermion-to-qubit mapping; NISQ

1 Introduction

Electronic-structure calculations play an important role in the predictive power of molecular simulations in chemistry and materials science. Among the explored properties, the potential-energy surfaces that map ground-state energy as a function of nuclear geometry are indispensable for quantifying bond-dissociation energetics, reaction barriers, and catalytic mechanisms [1, 2]. The H_2 and CO_2 molecules exemplify these challenges in the context of the global energy transition: H_2 production relies on optimised hydrogen-evolution catalysis, while CO_2 capture and conversion depend on understanding chemisorption and O–C–O bond activation [3, 4]. Standard wavefunction methods encounter well-documented difficulties in describing bond-breaking regions, where static correlation becomes dominant [5].

Quantum computing offers a fundamentally different route to solving the electronic-structure problem by encoding the fermionic Fock space directly in the Hilbert space of a qubit register. Superposition and entanglement allow the native representation of nonlocal correlations and antisymmetry, providing a pathway that avoids the exponential scaling of exact classical methods such as full configuration interaction (FCI) [6–9]. In the noisy intermediate-scale quantum (NISQ) era [10, 11], limited qubit counts, short coherence times, and gate errors restrict practical simulations to small molecular systems. Hybrid quantum–classical algorithms such as the variational quantum eigensolver (VQE) [12, 13], combined with dimension-reduction strategies—active-space selection [14], fermion-to-qubit mappings [15–17], and symmetry-informed qubit tapering [18–20]—make such calculations tractable on current hardware. VQE does not yet circumvent exponential scaling for systems beyond

trivial size; it provides approximate solutions whose accuracy and resource requirements must be carefully benchmarked against classical references [21, 22].

The H_2 molecule has long served as a minimal benchmark for quantum algorithms [23, 24] because its minimal basis (STO-3G) requires only four spin-orbitals. Carbon dioxide, by contrast, demands 30 spin-orbitals in the same basis, placing it near the boundary of NISQ capabilities. Prior quantum-computing studies on CO_2 addressed vibrational structure and anharmonic effects [25, 26], interactions with amines [21], and binding in metal–organic frameworks for carbon capture [3, 27, 28]. None of these works, however, computed the *electronic* potential-energy surface of isolated linear CO_2—particularly under symmetric O–C–O bond stretching, which directly probes the dissociation regime where static correlation is strongest.

In this work we employ VQE with a unitary coupled-cluster singles-and-doubles (UCCSD) ansatz to compute the ground-state PES for H_2 and CO_2 over a range of interatomic distances. The calculations on the H_2 molecule are used to calibrate the algorithm, establish noise-modelling baselines, and to validate results against FCI reference data. For CO_2, active-space reduction and symmetry-adapted qubit tapering bring the problem into a NISQ-compatible regime of 4–8 qubits. We systematically compare the circuit-depth and gate-count trade-offs of three fermion-to-qubit mappings—Jordan–Wigner (JW), Bravyi–Kitaev (BK), and parity—both with and without tapering, across expanding active spaces. Hardware-noise emulations using calibration data from IBM quantum processors quantify the energy penalties imposed by current devices. The resulting analysis provides a concrete entry point for integrating quantum-computing tools into materials-science electronic-structure workflows.

2 Computational methods

2.1 General framework

The electronic structures of neutral singlet H_2 and CO_2 were modelled within the Born–Oppenheimer approximation using VQE [12, 13]. All quantities are expressed in atomic units unless noted otherwise. The non-relativistic molecular Hamiltonian, the Born–Oppenheimer separation, the second-quantised representation, and the definitions of the one- and two-electron integrals are provided in the Supplementary Information (SI).

The fermionic Hamiltonian was mapped to qubit operators using the JW, BK, or parity transformation [15–17], followed by symmetry-informed qubit tapering (particle-number parity, S_z parity, and spin-flip symmetries) [18–20]. Resource counts (qubits and Pauli strings) for each mapping, before and after tapering, are reported in Tables 1–3 and in the SI.

The trial wavefunction was prepared with the UCCSD ansatz [29, 30],

$$|\psi(\boldsymbol{\theta})\rangle = e^{\hat{T}(\boldsymbol{\theta})-\hat{T}^\dagger(\boldsymbol{\theta})}|\phi_{\mathrm{HF}}\rangle, \tag{1}$$

where $|\phi_{\mathrm{HF}}\rangle$ is the Hartree–Fock (HF) reference and $\boldsymbol{\theta}$ are variational parameters optimised in a hybrid quantum–classical loop.

2.2 Software and optimisation

All simulations employed Qiskit Nature 0.7.2 [31] interfaced with PySCF 2.5.0 [32] for integrals and HF calculations, supplemented by PennyLane 0.35.0 and OpenFermion 1.6.0 [33]. The sequential least-squares programming (SLSQP) optimiser was selected after benchmarking against analytic quantum gradient descent (AQGD), COBYLA, Nelder–Mead, and simultaneous perturbation stochastic approximation (SPSA) using 4096-shot sampling (tolerance 10^{-8}, maximum 2000 iterations); SLSQP consistently provided the best convergence for the UCCSD ansatz (see SI for full benchmarking results). A fixed random seed ensured reproducibility of

all stochastic components. Complete software versions and library details are listed in the SI.

2.3 Simulation regimes

Three simulation regimes were employed:

1. *Exact statevector*—noiseless evaluation of expectation values.
2. *Shot-noise-limited*—4096 shots by default (100–7000 for the convergence study).
3. *Noisy emulation*—Qiskit Aer noise models calibrated to the IBM processors Kyoto (127 qubits), Osaka (127 Q), Brisbane (127 Q), Torino (133 Q), Jakarta (7 Q), and Manila (5 Q), incorporating T_1/T_2 relaxation, single- and two-qubit gate errors, and readout errors.

For shot-noise and noisy simulations, SPSA replaced SLSQP because gradient-based optimisers degrade under stochastic measurement noise.

2.4 Basis set and geometry scans

The minimal STO-3G basis set (Slater-type orbital expanded in three Gaussian primitives) [34, 35] was used for all VQE calculations. While STO-3G is quantitatively inadequate for production-level chemistry, it is the standard benchmark basis in quantum-computing studies because it minimises qubit counts. Potential-energy curves were obtained by scanning the H–H bond length from 0.3 Å to 2.5 Å (0.05 Å increments, 45 geometry points) and the symmetric C–O stretch of linear CO_2 from 0.8 Å to 2.5 Å (35 geometry points). The resulting discrete energies were fitted to a Morse potential via nonlinear least-squares regression to extract equilibrium bond lengths (r_e), dissociation energies (D_e), and harmonic frequencies (ω_e). Parameter uncertainties were estimated from the fit covariance matrix.

All energy differences reported below are computed with respect to the FCI reference in the same STO-3G basis unless stated otherwise. Comparisons with experimental spectroscopic values from the NIST Chemistry WebBook [36] are indicated explicitly. Classical benchmarks at the HF, density-functional theory (DFT/B3LYP), and FCI levels were performed with

PySCF using STO-3G. The hybrid quantum–classical workflow is illustrated in Fig. 1.

2.5 CO_2 active-space design

Active-space selection [14] was used to reduce the qubit requirements for CO_2. Three complete active spaces (CAS) were investigated:

- CAS(2e,2o): two electrons in two orbitals (minimal correlation);
- CAS(4e,4o): four electrons in four orbitals—$2\sigma_g$, one component of $1\pi_u$, one component of $1\pi_g^*$, and $3\sigma_u$—preserving the dominant valence bonding/antibonding character;
- CAS(6e,6o): six electrons in six orbitals, adding $2\sigma_u$ and $3\sigma_g$.

Orbitals were selected according to proximity to the Fermi level; complete degenerate π pairs were always included together. Full orbital compositions and energies are given in the SI.

For the initial active-space comparison (Table 1), the JW mapping was used without tapering to isolate the effect of active-space size. In subsequent analyses, all three mappings (JW, BK, parity) were compared both without tapering (Table 2) and with full symmetry tapering (Table 3).

3 Results

3.1 H_2 molecule

3.1.1 Exact and sampling regimes

Figure 2 presents the PES of H_2 computed with classical reference methods (HF, DFT/B3LYP, and FCI) alongside the statevector VQE and finite-shot VQE results, all in the STO-3G basis. As expected, the statevector UCCSD-VQE on this two-electron, four-spin-orbital system is algebraically equivalent to FCI: UCCSD includes all possible excitations in this minimal space, and the two curves overlap to within numerical precision ($\sim 10^{-7}$ Ha) across the entire scanned range. At the equilibrium distance ($r_e \approx 0.734$ Å), the statevector VQE

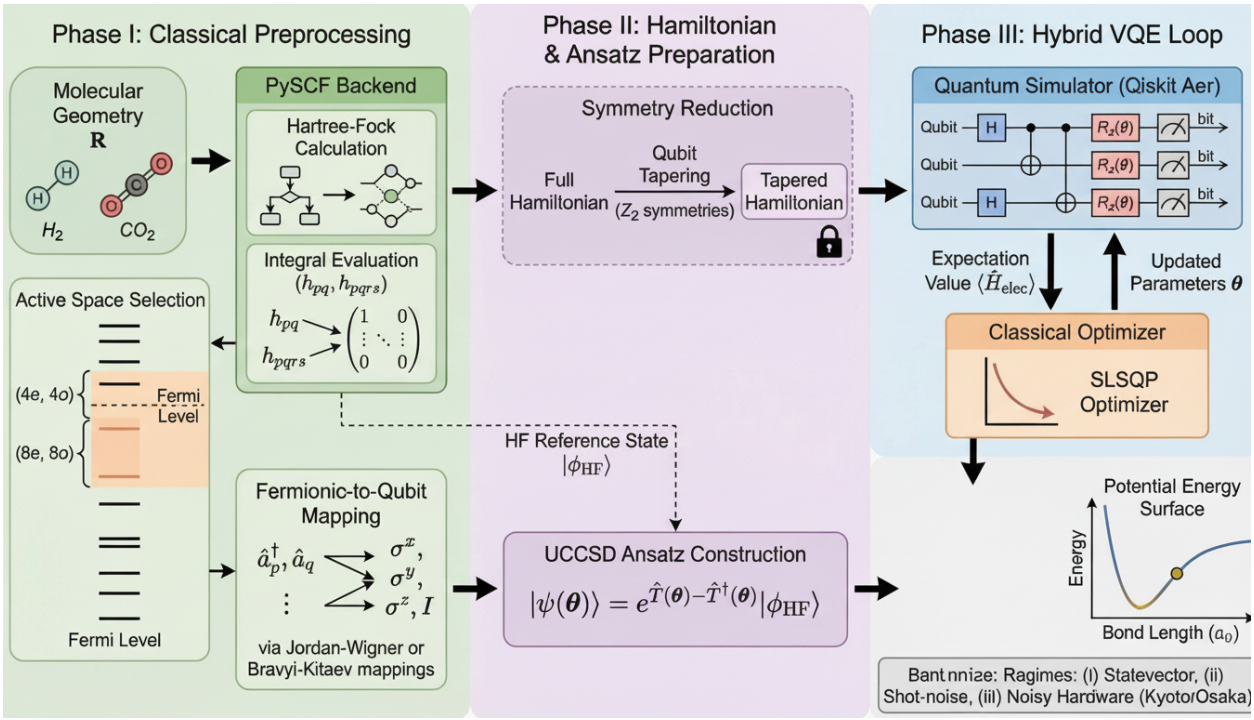


Figure 1: Overview of the hybrid quantum–classical VQE workflow. Phase I: classical preprocessing with PySCF (integrals, HF reference, active-space selection). Phase II: Hamiltonian construction, fermion-to-qubit mapping, and symmetry tapering. Phase III: VQE optimisation loop on the quantum simulator, producing the ground-state energy at each nuclear geometry. Three simulation regimes were employed: (i) exact statevector, (ii) shot-noise-limited, and (iii) noisy hardware emulation.

yields $E_0 = -1.137\,\text{Ha}$.

Under finite-shot sampling (4096 shots), the VQE curve retains the correct shape with small deviations arising from statistical noise (Fig. 2c). A comprehensive survey comparing quantum-chemistry libraries (Qiskit Nature, PennyLane, OpenFermion/Cirq), classical optimisers, and basis sets confirmed the robustness of the pipeline; full details and figures are provided in the SI.

3.1.2 Shot-count convergence

To quantify the statistical precision required, the number of shots was varied from 100 to 7000 using the SPSA optimizer (Fig. 3). Minimum energies extracted from Morse potential fits to the equilibrium region achieve chemical accuracy at 1550 shots and stabilize between 3000 and 4000 shots, converging to $-1.1384\,\text{Ha}$ ($\sigma = 0.58\,\text{mHa}$)—within 1.1 mHa of the STO-3G FCI reference ($-1.1373\,\text{Ha}$). Given that the stability is reached beyond 4000 shots, all subsequent sampling simulations utilize 4096 shots according to this precision threshold.

3.1.3 Noise-model emulations

Realistic noise models derived from the calibration data of six IBM quantum processors (5–133 qubits) were applied (Fig. 4). The noisy PES curves retain the qualitative shape of the ideal surface but are shifted upward by 0.2 Ha to 1.5 Ha, depending on the device profile and the bond length. The magnitude of this shift far exceeds the chemical-accuracy threshold ($\sim 1.6\,\text{mHa}$) and arises from accumulated gate errors in the UCCSD circuit. The shift is largest at short bond lengths, where the circuit explores a wider parameter space. These results quantify the noise floor that error-mitigation strategies must overcome for near-term hardware implementations.

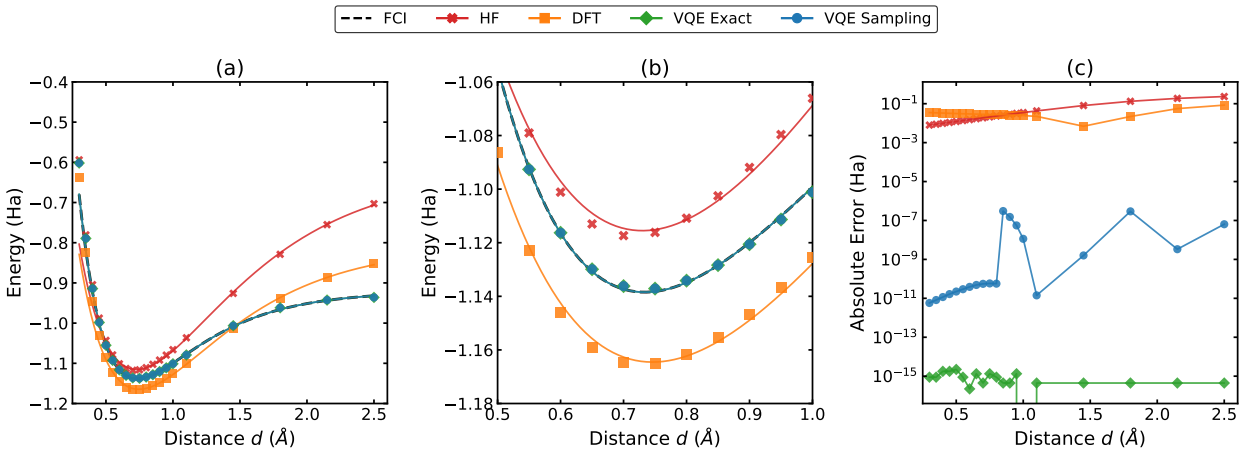


Figure 2: H_2 potential-energy surface in the STO-3G basis. (a) Full PES comparing FCI (dashed black), HF (red), DFT/B3LYP (orange), statevector VQE (green), and 4096-shot VQE (blue). (b) Zoom near equilibrium showing the VQE–FCI overlap and the HF/DFT deviations. (c) Absolute error relative to FCI on a logarithmic scale; the statevector VQE matches FCI to $\sim 10^{-15}$ Ha (machine precision), while shot-noise errors remain below 10^{-3} Ha near equilibrium.

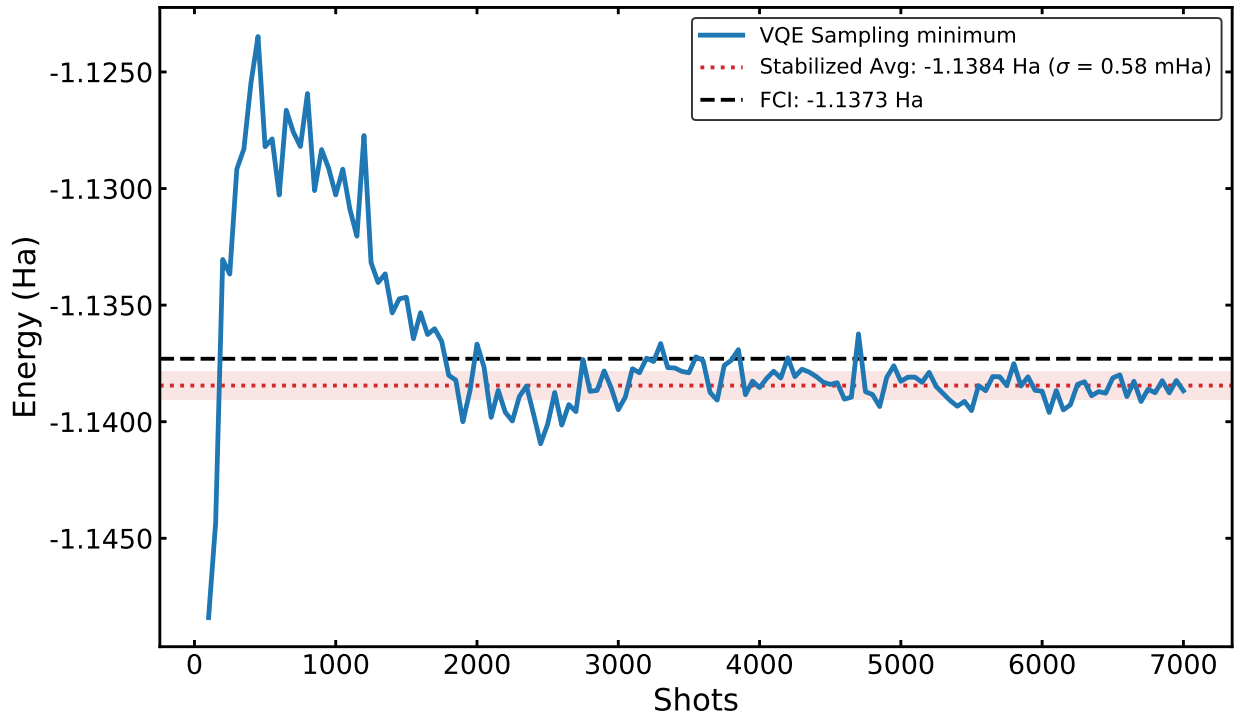


Figure 3: Convergence of the fitted minimum energy of H_2 as a function of shot count.The solid blue line traces the Morse fitted VQE minima, achieving chemical accuracy at 1,550 shots. Beyond 4,000 shots, the energy stabilizes (dotted red line) with a standard deviation of 0.58 mHa (shaded region), sitting marginally below the exact FCI limit (dashed black line)

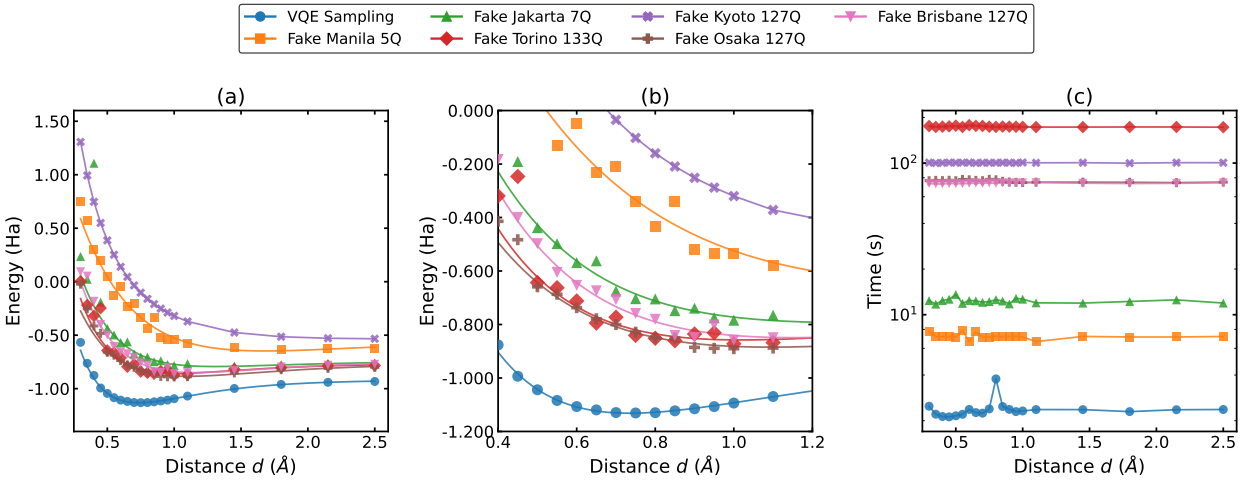


Figure 4: Effect of hardware-noise emulation on the H_2 PES. (a) Full PES under noise models from six IBM devices (5–133 qubits) compared with the ideal 4096-shot VQE (blue circles). (b) Zoom near equilibrium showing the noise-induced upward energy shift. (c) Wall-clock time per geometry point; noisy simulations are 10–100× slower than ideal sampling due to the noise-channel overhead.

3.2 CO_2 molecule

3.2.1 Active-space comparison

Table 1 summarises the equilibrium energies, circuit resources, and geometries for the three active spaces investigated under the JW mapping without tapering. The circuit depth and controlled-NOT (CNOT) gate count grow rapidly with active-space size: from 56 CNOTs for CAS(2e,2o) to 10 344 for CAS(6e,6o). The CAS(4e,4o) model recovers 32% of the electron-correlation energy ($[E_{\mathrm{CAS}} - E_{\mathrm{HF}}]/[E_{\mathrm{FCI}} - E_{\mathrm{HF}}] = 0.073/0.227$) for 1440 CNOTs, while expanding to CAS(6e,6o) gains only an additional 10 percentage points at a sevenfold increase in gate count. CAS(4e,4o) therefore represents the best trade-off between correlation recovery and circuit depth. The CNOT gate count is monitored as a proxy for circuit complexity, as two-qubit entangling gates like CNOT exhibit the highest error rates and longest execution times, dominating overall circuit fidelity and runtime [37].

The corresponding PES are shown in Fig. 5. All VQE curves lie between the uncorrelated HF baseline (−185.067 Ha) and the FCI benchmark (−185.294 Ha), with progressive improvement as the active space expands. The CAS(4e,4o) equilibrium bond length is 1.171 Å. This

value is closer to the experimental C–O distance (1.162 Å) than the FCI/STO-3G prediction (1.222 Å), an effect attributable to partial cancellation between the basis-set error (STO-3G overbinds) and the active-space truncation error.

Table 1: Equilibrium energies and circuit metrics for CO_2 active-space comparison using the JW mapping without tapering (exact statevector VQE, STO-3G basis).

Method	**Space**	**Qubits**	**Depth**	**CNOTs**	$d_{\min}$ (Å)	$E_{\min}$ (Ha)
HF	Full	–	–	–	1.177	−185.067
VQE	CAS(2,2)	4	73	56	1.173	−185.081
VQE	CAS(4,4)	8	1711	1440	1.171	−185.140
VQE	CAS(6,6)	12	11803	10344	1.174	−185.162
FCI	Full	–	–	–	1.222	−185.294

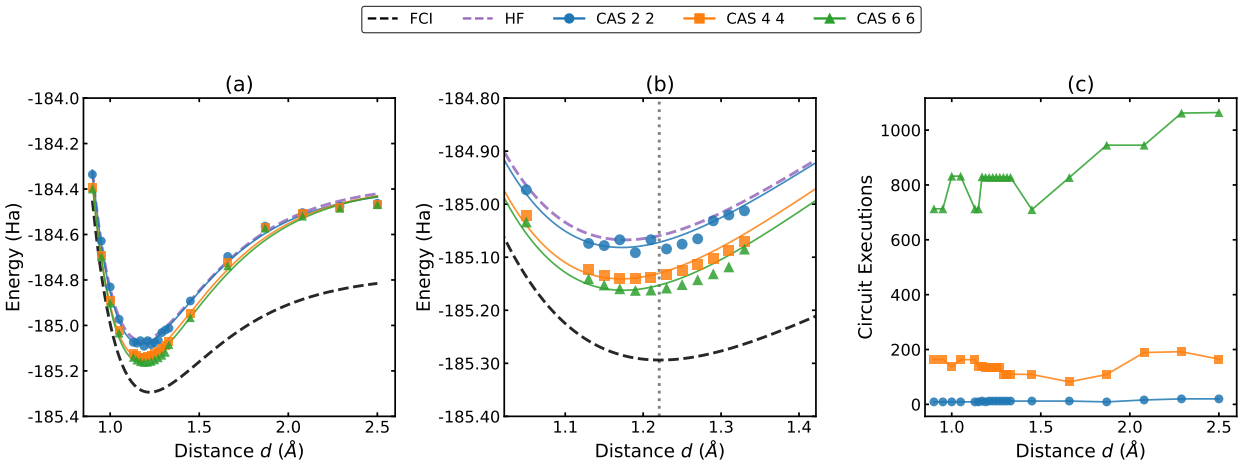


Figure 5: CO_2 potential-energy surface under symmetric C–O bond stretching (STO-3G basis, exact statevector VQE). (a) Full PES for HF (dashed purple), FCI (dashed black), and three CAS sizes. (b) Zoom near equilibrium; the vertical dotted line marks the experimental C–O distance (1.162 Å). (c) Number of circuit executions (VQE iterations) at each geometry point.

3.2.2 Fermion-to-qubit mapping comparison

Table 2 compares the JW, BK, and parity mappings across all three active spaces without tapering. Because the eigenvalues are invariant under unitary mapping transformations, the PES from all three mappings overlap for each CAS; minor numerical deviations (∼8 mHa)

occasionally appear in minimal active spaces due to the classical optimiser navigating different parameter landscapes.

The parity mapping yields the lowest resource requirements in all configurations. For CAS(2e,2o), parity reduces the CNOT count from 56 (JW) to 4—a 93% reduction—because the two-qubit parity encoding compresses the Hamiltonian structure. For larger active spaces this relative advantage diminishes: the reduction is 24% for CAS(4e,4o) and 11% for CAS(6e,6o). This trend arises because the UCCSD ansatz generates an exponentially growing number of double-excitation operators in larger spaces, limiting the scope for mapping-level compression.

The BK mapping provides a negligible 1.1% CNOT reduction for CAS(6e,6o) relative to JW. This outcome is consistent with theoretical expectations: BK's advantage scales as $O(\log n)$ versus JW's $O(n)$ for the locality of Pauli strings, but this scaling difference becomes significant only for systems with $n \gtrsim 19$ qubits [38]. For the 4–12 qubit circuits studied here, the logarithmic advantage is negligible.

Table 2: Comparison of fermion-to-qubit mappings across CO_2 active spaces (no tapering, exact statevector VQE, STO-3G).

Mapper	**CAS**	**Qubits**	**Params.**	**Depth**	**CNOTs**	**Time (s)**
Bravyi–Kitaev	(2,2)	4	3	59	38	0.35
Jordan–Wigner	(2,2)	4	3	73	56	0.40
Parity	(2,2)	2	3	11	4	0.30
Bravyi–Kitaev	(4,4)	8	26	1612	1316	9.55
Jordan–Wigner	(4,4)	8	26	1711	1440	11.53
Parity	(4,4)	6	26	1359	1096	8.55
Bravyi–Kitaev	(6,6)	12	117	11784	10228	462.69
Jordan–Wigner	(6,6)	12	117	11803	10344	502.02
Parity	(6,6)	10	117	10672	9200	310.90

3.2.3 Effect of symmetry tapering

Table 3 presents the circuit metrics after applying full $\mathbb{Z}_2$ symmetry tapering. Because tapering is an exact mathematical projection, it incurs no loss of chemical accuracy. For

CAS(2e,2o), tapering eliminates all entanglement requirements, compressing the simulation to a single qubit with zero CNOT gates. For CAS(4e,4o) and CAS(6e,6o), tapering equalises the qubit count across all three mappings (5 and 8 qubits, respectively).

After tapering, JW compiles more efficiently than parity or BK: CAS(4e,4o) requires only 734 CNOTs under JW versus 792 for both BK and parity; CAS(6e,6o) requires 3036 (JW) versus 3378 (BK) and 3724 (parity). The relative error of the VQE ground-state energy remains below 0.1% of the FCI limit even at maximum compression, validating the fidelity of the active-space approximation.

The combination of JW mapping, symmetry tapering, and CAS(4e,4o)—yielding 5 qubits, 26 variational parameters, and 734 CNOTs—provides the optimal framework for simulating the CO_2 PES on near-term hardware.

Table 3: Tapered mapper comparison for CO_2 (exact statevector VQE, STO-3G). Δd is the relative deviation of the equilibrium bond length from experiment; ΔE is the relative energy error versus FCI.

Mapper	**CAS**	**Qubits**	**Depth**	**CNOTs**	Δd (%)	ΔE (%)
Bravyi–Kitaev	(2,2)	1	2	0	−3.94	0.116
Jordan–Wigner	(2,2)	1	1	0	−3.73	0.120
Parity	(2,2)	1	1	0	−4.04	0.117
Bravyi–Kitaev	(4,4)	5	1063	792	−4.11	0.083
Jordan–Wigner	(4,4)	5	1004	734	−4.11	0.083
Parity	(4,4)	5	1065	792	−4.11	0.083
Bravyi–Kitaev	(6,6)	8	4094	3378	−3.87	0.071
Jordan–Wigner	(6,6)	8	3776	3036	−3.87	0.071
Parity	(6,6)	8	4485	3724	−3.87	0.071

3.2.4 Noise-model emulation for CO_2

Figure 6 presents the tapered 5-qubit CAS(4e,4o) PES under finite sampling and hardware-noise emulation. Under 4096-shot sampling (SPSA optimiser), the VQE curve shows increased scatter relative to the exact statevector baseline, with the ground-state minimum shifting upward to −184.370 Ha—a 0.77 Ha deviation from the exact CAS(4e,4o) value. The fitted

equilibrium bond length under sampling (1.169 Å) remains close to the exact CAS(4e,4o) result (1.171 Å).

When subjected to the IBM Kyoto noise model, the energy minimum degrades further to −183.855 Ha (a 1.29 Ha upward shift) and the equilibrium distance stretches to 1.264 Å. The absolute-error analysis (Fig. 6b) reveals that the noise-induced penalty is not strictly constant: it ranges from approximately 0.1 Ha near equilibrium to several hartree at compressed geometries. Despite this significant degradation, the global minimum remains identifiable on the noisy PES, suggesting that the *topology* of the surface is preserved even without error mitigation.

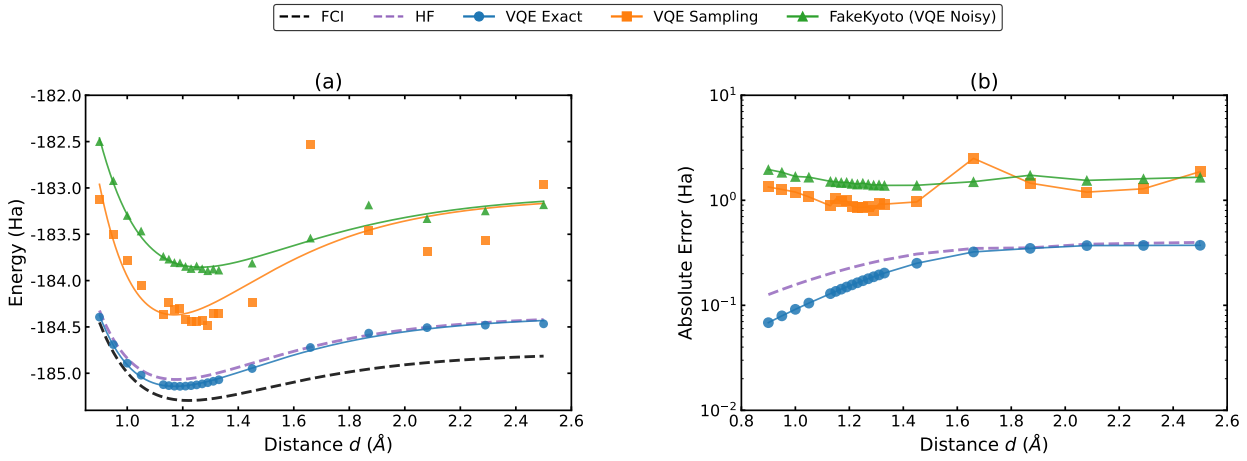


Figure 6: Effect of sampling noise and hardware-noise emulation on the CO_2 CAS(4e,4o) PES (tapered JW, 5 qubits). (a) PES comparing exact statevector VQE (blue), 4096-shot sampling (orange), and Kyoto noise-model emulation (green) against the FCI (dashed black) and HF (dashed purple) references. (b) Absolute energy error relative to FCI; the sampling error is ∼0.8 Ha at the minimum and the noise-model error reaches 1.3 Ha.

4 Discussion

4.1 VQE accuracy and the role of active-space selection

The central finding of this work is that VQE with a UCCSD ansatz, combined with active-space reduction and symmetry tapering, can reproduce the electronic PES of CO_2 within a compact

5-qubit circuit while capturing a chemically meaningful fraction of the electron-correlation energy. The CAS(4e,4o) model recovers 32% of the correlation energy for 734 CNOTs, whereas expanding to CAS(6e,6o) recovers 42% at a fourfold increase in gate count (3036 CNOTs). This diminishing-returns behaviour is expected: the additional orbitals in CAS(6e,6o) contribute progressively smaller correlation increments, while the UCCSD circuit depth grows combinatorially with the number of active orbitals.

A notable observation is that the VQE/CAS(4e,4o) equilibrium bond length (1.171 Å) lies closer to the experimental value (1.162 Å) than the FCI/STO-3G prediction (1.222 Å). This is attributable to fortuitous error cancellation: the STO-3G basis overestimates the bond length, while the active-space truncation partially compensates this error by omitting correlation contributions that further stabilise the dissociated limit. Users of small-basis VQE calculations should be cautious about interpreting such agreement as evidence of chemical accuracy; it is the *relative* performance against the same-basis FCI reference that is physically meaningful.

4.2 Mapping and tapering trade-offs

The systematic comparison of three fermion-to-qubit mappings provides practical guidance for VQE practitioners. Without tapering, the parity mapping offers the lowest CNOT counts across all active spaces. However, after symmetry tapering—which should always be applied when molecular symmetries are available—the JW mapping compiles most efficiently, requiring 8% fewer CNOTs than BK or parity for CAS(4e,4o). This result reflects the structure of the tapered Hamiltonian: JW's simple string encoding aligns well with the reduced symmetry sector, whereas BK's tree-structured encoding creates overhead in the compiler's routing step.

For the small qubit counts studied here (1–12), BK's theoretical $O(\log n)$ advantage over JW [16] is negligible. These findings are consistent with a concurrent resource-estimation study by Anurag et al., who benchmarked the same three mappings with tapering on 13 small

molecules and reported qualitatively similar trends [39]. The practical implication is that the choice of mapping matters less than the application of tapering; for systems requiring fewer than ∼20 qubits, JW with full tapering is a robust default choice.

4.3 Noise and the path to real-hardware deployment

The noise-model emulations reveal that unmitigated hardware errors introduce energy penalties of 0.2 Ha to 1.5 Ha for H_2 (1 qubit) and 0.8 Ha to 1.3 Ha for CO_2/CAS(4e,4o) (5 qubits), far exceeding the chemical-accuracy threshold. These errors arise primarily from two-qubit gate infidelity accumulated over the hundreds of CNOT layers in the UCCSD circuit.

Despite these large absolute errors, the topology of the PES is preserved: the global minimum remains identifiable, and the qualitative shape of the dissociation curve is maintained. This observation suggests that error-mitigation techniques such as zero-noise extrapolation (ZNE) [40] or Clifford data regression (CDR) [41]—which correct systematic biases while preserving curve topology—may be effective for recovering quantitative accuracy from noisy VQE data.

We emphasise that all results reported here are classical emulations of noisy circuits, not measurements on physical quantum hardware. Recent work has demonstrated that VQE can achieve chemical accuracy for small molecules on real IBM devices when combined with appropriate error mitigation [23, 28]. Extending the present CO_2 study to physical hardware and applying error mitigation is a natural next step.

4.4 Limitations

Several limitations should be acknowledged. First, the STO-3G basis is too small for quantitative chemistry; it serves only as a benchmark for algorithmic performance. Calculations with larger basis sets (e.g., cc-pVDZ) would increase qubit counts substantially and require more aggressive active-space or embedding strategies. Second, UCCSD is not the most hardware-efficient ansatz: adaptive approaches such as ADAPT-VQE have been shown to produce

shorter circuits for similar accuracy [28, 42]. Third, only symmetric C–O stretching was explored; asymmetric stretching and bending modes—relevant for vibrational spectroscopy and reaction dynamics—remain for future investigation. Finally, the absence of real-hardware results and error mitigation limits the conclusions that can be drawn about practical NISQ deployment.

5 Conclusions

We have presented a systematic VQE study of the ground-state PES for H_2 and CO_2, mapping the algorithmic and resource requirements from the simplest benchmark to a polyatomic molecule. For CO_2, the combination of CAS(4e,4o) active-space reduction, JW mapping, and symmetry tapering compresses the problem to 5 qubits and 734 CNOT gates while recovering 32% of the correlation energy at less than 0.1% deviation from the FCI limit. A systematic comparison of three fermion-to-qubit mappings shows that, after tapering, JW compiles most efficiently for these small systems—a finding consistent with BK's theoretical advantage manifesting only at larger qubit counts. Noise-model emulations quantify the energy penalties of current IBM hardware profiles and demonstrate that the PES topology is preserved even under significant noise, providing a concrete baseline for error-mitigation strategies.

This work provides a practical reference for materials-science researchers entering the quantum-computing space: the mapping, tapering, and active-space choices benchmarked here are directly transferable to analogous molecular systems relevant to catalysis and carbon capture. Future work will focus on implementing ZNE and CDR error mitigation on physical quantum devices, exploring adaptive ansatz designs (ADAPT-VQE), and extending to larger basis sets and active spaces as hardware capabilities advance.

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Author contributions

A.T. initiated the project and contributed to the conceptual development of the quantum-computing framework. L.L. and G.F.T. implemented the codes, performed the quantum simulations for H_2 and CO_2, and carried out the data analysis. A.K. orchestrated the overall research, organised the manuscript structure, and led the writing of the original draft. C.R.M. supervised the project. All authors discussed the results and contributed to the review and editing of the final manuscript.

Declaration of AI use

Artificial intelligence tools (ChatGPT, OpenAI; Claude, Anthropic) were used to improve the clarity and grammar of the manuscript text and to assist in literature searches. All scientific content, interpretations, and conclusions are the sole responsibility of the authors.

Competing interests

The authors declare no competing interests.

Data availability

All data underlying this study will be made available in a public repository upon acceptance.

Code availability

All simulation code will be made available in a public repository upon acceptance.

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